

```

a = input()
vowels = []
consonants = []
for i in a:
    if i == "a" or i == "e" or i == "i" or i == "o" or i == "u" or i == "A" or
    i == "E" or i == "I" or i == "O" or i == "U":
        if i not in vowels:
            vowels.append(i)
    else:
        if i not in consonants:
            consonants.append(i)
print("Vowels ", vowels)
print("Consonants ", consonants)

```

```

haneesh
Vowels  ['a', 'e']
Consonants  ['h', 'n', 's']

```

```

#reverse a string

a = "abcdefghijklmnopqrstuvwxyz"
print(a[::-1] + a)

for i in range(len(a)-1, -1, -1):
    print(a[i], end = "")

```

```

zyxwvutsrqponmlkjihgfedcbaabcdefghijklmnopqrstuvwxyz
zyxwvutsrqponmlkjihgfedcba

```

```

word1 = input("Enter first word: ")
word2 = input("Enter second word: ")

if sorted(word1) == sorted(word2) and len(word1) == len(word2):
    is_anagram = True
else:
    is_anagram = False

print("Are they anagrams?", is_anagram)

```

```

Enter first word: silent
Enter second word: listen
Are they anagrams? True

```

```

#48 to 57 digits 0 to 9
#65 to 90 A to Z
#97 to 122 a to z

enter = input("Enter: ")
print(ord(enter))

```

```

Enter: Z
90

```

```

num = int(input("Enter number: "))
order = len(str(num))
total = 0
temp = num

while temp > 0:
    total += (temp % 10) ** order
    temp //= 10

if num == total:
    print("Armstrong number")
else:
    print("Not an armstrong number")

```

```

Enter number: 1624
Not an armstrong number

```

```
#prime numbers in a range

num1 = int(input("Enter first number: "))
num2 = int(input("Enter second number: "))
for i in range(num1, num2+1):
    for j in range(2, i):
        if i % j == 0:
            break
    else:
        print(i)
```

```
Enter first number: 2
Enter second number: 10
2
3
5
7
```

```
#+,-,0
num = int(input("Enter: "))
if num > 0:
    print("Positive")
elif num < 0:
    print("Negative")
else:
    print("Zero")
```

```
Enter: 4
Positive
```

```
#eligibility to vote, above 18, area

age = int(input("Enter age: "))
area = input("Enter area: ")
if age >= 18 and area == "vij":
    print("Eligible to vote")
else:
    print("Not eligible to vote")
```

```
Enter age: 22
Enter area: vij
Eligible to vote
```

```
#leap year

year = int(input("Enter year: "))
if year % 4 == 0 and year % 100 != 0:
    if year % 400 == 0:
        print("Leap year")
else:
    print("Not a leap year")
```

```
Enter year: 1999
Not a leap year
```

```
#simple calculator

num1 = int(input("Enter first number: "))
num2 = int(input("Enter second number: "))
op = input("Enter operator: ")
if op == "+":
    print(num1 + num2)
elif op == "-":
    print(num1 - num2)
elif op == "*":
    print(num1 * num2)
elif op == "/":
    print(num1 / num2)
else:
    print("Invalid operator")\
```

```
Enter first number: 3
Enter second number: 4
Enter operator: -
-1
```

```
#perfect number
per = int(input("Enter number: "))
sum = 0
for i in range(1, per):
    if per % i == 0:
        sum += i
if sum == per:
    print("Perfect number")
else:
    print("Not a perfect number")
```

Enter number: 12
Not a perfect number

```
#using while loop count no of even digits in a number
```

```
num = int(input("Enter number: "))
count = 0
while num > 0:
    if num % 2 == 0:
        count += 1
    num //= 10
print(count)
```

Enter number: 24680
5

```
freq = {1:"Apple", 2:"Banana", 3:"Cherry"}
print(freq)
print(freq.keys())
print(freq.values())
print(freq.items())
print(freq.get(1))
print(freq.get(4))
```

```
{1: 'Apple', 2: 'Banana', 3: 'Cherry'}
dict_keys([1, 2, 3])
dict_values(['Apple', 'Banana', 'Cherry'])
dict_items([(1, 'Apple'), (2, 'Banana'), (3, 'Cherry')])
Apple
None
```

```
word = "Chess"
freq = {}
for char in word:
    if char in freq:
        freq[char] += 1
    else:
        freq[char] = 1
print(freq)
```

{'C': 1, 'h': 1, 'e': 1, 's': 2}

```
inp = "aabbccccc"
freq = {}
for char in inp:
    if char in freq:
        freq[char] += 1
    else:
        freq[char] = 1
print(freq)
```

{'a': 2, 'b': 3, 'c': 4}

```
a = [9, 5, 3, 4, 5, 3, 9, 4]
b = []
```

```
for i in a:
    if i not in b:
        b.append(i)

print(sorted(b))
```

[3, 4, 5, 9]

```
inp = "aabbccccc"
freq = {}
for char in inp:
    if char in freq:
        freq[char] += 1
    else:
        freq[char] = 1
for char in freq:
    print(char, freq[char], sep=" ", end=" ")
```

a2b3c4

```
s1 = input("Enter even number of String only example is abcd: ")
s2 = input("Enter even number of String only example is cdab: ")
if s2 == s1[2:] + s1[:2]:
    print("True")
else:
    print("False")

print(s1[2:]+s1[:2])
```

Enter even number of String only example is abcd: abcd
Enter even number of String only example is cdab: cdab
True
cdab

```
s = "A man, a plan, A canal, panama"
cleaned = s.replace(", ", "").replace(" ", "").lower()
print(cleaned)
s1 = cleaned[::-1]
if cleaned == s1:
    print("Palindrome")
else:
    print("Not a palindrome")
```

amanaplanacanalpanama
Palindrome

```
a = "amanaplanacanalpanama"
for i in range(len(a)-1):
    if a[i] == a[-i]:
        print("Palindrome")
        break
else:
    print("Not a palindrome")
```

Palindrome

```
#longest word in the sentence

sentence = input("Enter sentence: ")
words = sentence.split()
longest = ""
for word in words:
    if len(word) > len(longest):
        longest = word
print(longest)
```

Enter sentence: colab is the best
colab

```
# Simple solution using max()
sentence = input("Enter sentence: ")
words = sentence.split()

longest = max(words, key=len)
print(longest)
```

```
# Fixed version using the reversed function
sentence = input("Enter Sentence: ")
words = sentence.split()
for word in reversed(words):
    print(word, end=" ")
```

Enter Sentence: a b c d
d c b a